Question 6 - Feature flags, load testing and Selenium

Question 6: Implement feature flags, perform load testing, and demonstrate Selenium-based automated testing; record commands, configurations, screenshots, and observations.

All commands below were run on the lab machine and their output is copied verbatim from the recorded logs in `evidence/logs/`. Repository: https://github.com/ksamirdev/ensp461-devops-lab3.

Unit tests

```
$ /home/samir/lab-tools/bin/python -m pytest -v -p no:cacheprovider 2>&1 | tail -16
configfile: pytest.ini
testpaths: tests
plugins: platformdirs-4.14.5
collecting ... collected 9 items

tests/test_web.py::test_healthz_reports_healthy PASSED                   [ 11%]
tests/test_web.py::test_text_stats_counts_unique_words_case_insensitively PASSED [ 22%]
tests/test_web.py::test_stats_endpoint_returns_counts PASSED             [ 33%]
tests/test_web.py::test_stats_endpoint_rejects_invalid_text[body0] PASSED [ 44%]
tests/test_web.py::test_stats_endpoint_rejects_invalid_text[body1] PASSED [ 55%]
tests/test_web.py::test_metrics_exposes_request_counter PASSED           [ 66%]
tests/test_web.py::test_unknown_urls_share_one_metric_label PASSED       [ 77%]
tests/test_web.py::test_top_words_hidden_when_flag_off PASSED            [ 88%]
tests/test_web.py::test_top_words_served_when_flag_on PASSED             [100%]

============================== 9 passed in 0.10s ===============================
```

Feature flag

service/web.py (flag)

```
FEATURE_TOP_WORDS = os.environ.get("FEATURE_TOP_WORDS", "false").lower() == "true"
```

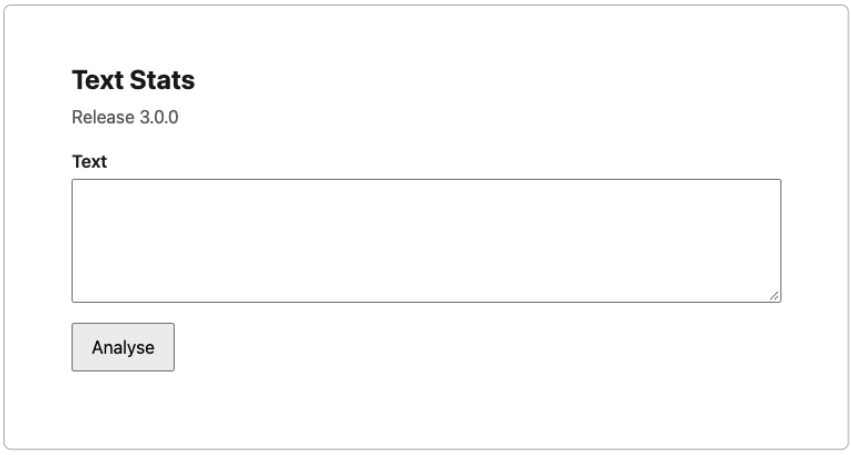
service/web.py (gated endpoint)

```
@app.post("/stats/top")
def stats_top():
    if not FEATURE_TOP_WORDS:
        return jsonify(error="feature 'top_words' is disabled"), 404
    text = read_text()
    if text is None:
        return jsonify(error="JSON field 'text' must be a string"), 400
    return jsonify(top_words=top_words(text))
```

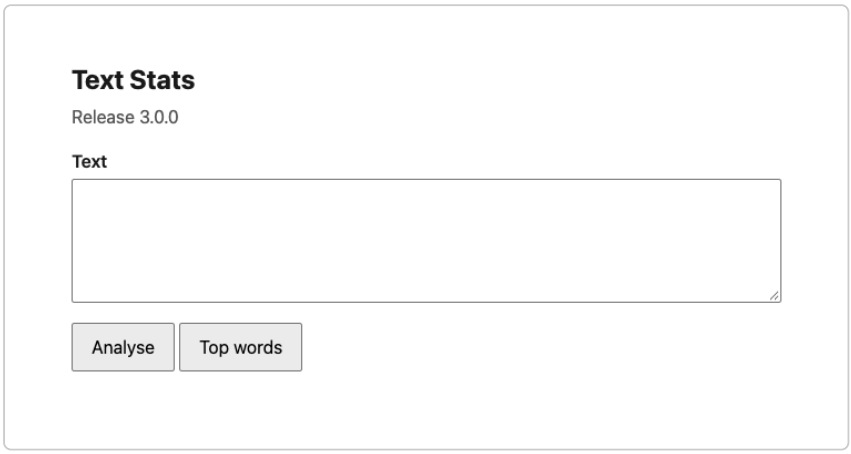
The same image is started twice. Only the environment variable differs.

```
$ docker run -d --name flag-off --network lab3-e2e -p 127.0.0.1:8320:5000 -e FEATURE_TOP_WORDS=false
text-stats:3.0.0
206d98f9d9bef7e7488e17eb1e08d7ae478abf755772544fc3fda6ca2dc81486
$ docker run -d --name flag-on --network lab3-e2e -p 127.0.0.1:8321:5000 -e FEATURE_TOP_WORDS=true t
ext-stats:3.0.0
2eb9933f1f55b805a4d1fcc8251aaab66a3bb62bff98b791fcc40837566b4c9c
$ curl -s -w ' HTTP %{http_code}\n' -H 'Content-Type: application/json' -d '{"text":"a a b"}' htt
p://localhost:8320/stats/top
{"error":"feature 'top_words' is disabled"}
 HTTP 404
$ curl -s -w ' HTTP %{http_code}\n' -H 'Content-Type: application/json' -d '{"text":"a a b"}' htt
p://localhost:8321/stats/top
```

```
{"top_words":[{"count":2,"word":"a"},{"count":1,"word":"b"}]}
 HTTP 200
```



Flag off: only the Analyse button is rendered



Flag on: the Top words button appears

Load test (ApacheBench)

```
$ ab -q -n 5000 -c 50 http://localhost:8300/healthz | grep -E 'Concurrency|Complete requests|Failed|
Requests per second|Time per request|99%|100%'
Concurrency Level:      50
Complete requests:      5000
Failed requests:        0
Requests per second:    4002.62 [#/sec] (mean)
Time per request:       12.492 [ms] (mean)
Time per request:       0.250 [ms] (mean, across all concurrent requests)
  99%     23
 100%     26 (longest request)
$ ab -q -n 5000 -c 50 -p /tmp/body.json -T application/json http://localhost:8300/stats | grep -E 'C
oncurrency|Complete requests|Failed|Non-2xx|Requests per second|Time per request|99%|100%'
Concurrency Level:      50
Complete requests:      5000
Failed requests:        0
Requests per second:    3574.92 [#/sec] (mean)
Time per request:       13.986 [ms] (mean)
Time per request:       0.280 [ms] (mean, across all concurrent requests)
  99%     30
 100%     46 (longest request)
$ ab -q -n 20000 -c 200 http://localhost:8300/healthz | grep -E 'Concurrency|Complete requests|Faile
d|Non-2xx|Requests per second|99%|100%'
Concurrency Level:      200
Complete requests:      20000
Failed requests:        0
Requests per second:    2995.98 [#/sec] (mean)
```

```
  99%    110
 100%    125 (longest request)
```



Grafana during the load tests. The Q6 script ran twice (see the bug below), so each /healthz and /stats burst appears twice.

Selenium browser test

e2e/test_ui_selenium.py (checks)

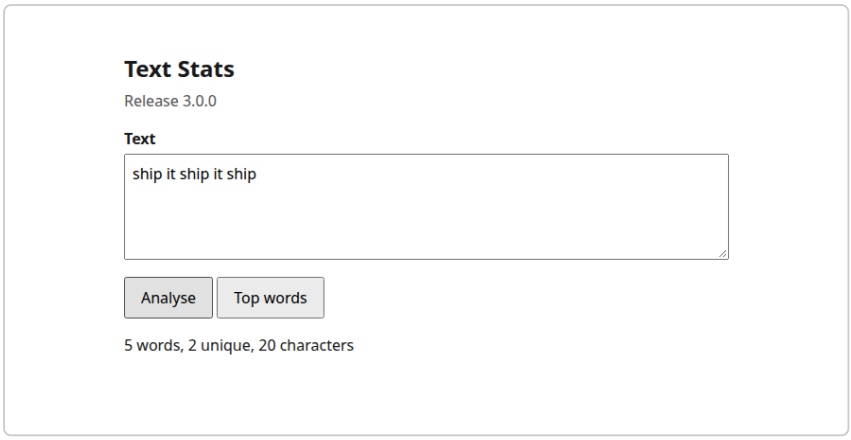
```
try:
    driver.get(f"{app_url}/ui")
    assert driver.title == "Text Stats"
    print("PASS page title is 'Text Stats'")

    driver.find_element(By.ID, "text").send_keys("ship it ship it ship")
    driver.find_element(By.ID, "analyse").click()
    WebDriverWait(driver, 5).until(result_contains("5 words, 2 unique, 20 characters"))
    print("PASS analyse shows '5 words, 2 unique, 20 characters'")
    if shots:
        driver.save_screenshot(f"{shots}/q6-selenium-analyse.png")

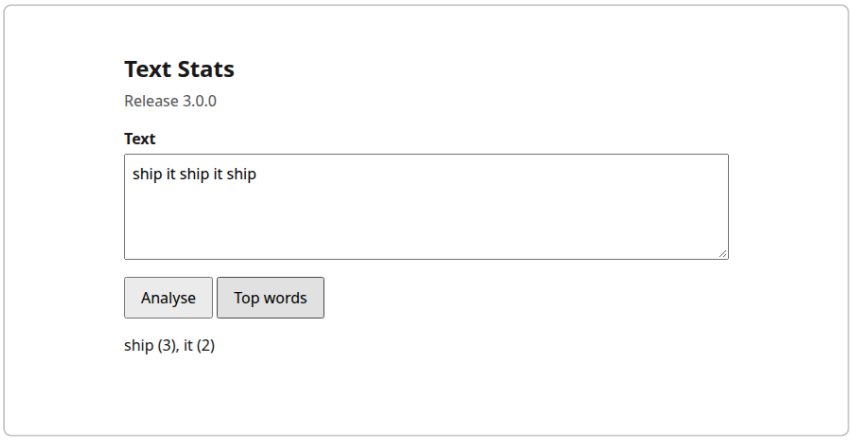
    driver.find_element(By.ID, "top").click()
    WebDriverWait(driver, 5).until(result_contains("ship (3), it (2)"))
    print("PASS top words shows 'ship (3), it (2)' with the flag on")
    if shots:
        driver.save_screenshot(f"{shots}/q6-selenium-top-words.png")
finally:
    driver.quit()
print("All browser checks passed")
```

```
$ docker run -d --name selenium --network lab3-e2e --shm-size=1g -p 127.0.0.1:4444:4444 mirror.gcr.io/selenium/standalone-chromium:latest
990c3960b9a1d48e57daaa999314656bdc344e1c5b36b93e27b0bf9139b4da82
$ until curl -sf http://localhost:4444/status | jq -e .value.ready >/dev/null; do sleep 2; done; curl -s http://localhost:4444/status | jq -c '{ready:.value.ready}'
{"ready":true}
$ /home/samir/lab-tools/bin/python e2e/test_ui_selenium.py http://flag-on:5000 http://localhost:4444 evidence/screenshots
PASS page title is 'Text Stats'
```

```
PASS analyse shows '5 words, 2 unique, 20 characters'
PASS top words shows 'ship (3), it (2)' with the flag on
All browser checks passed
```



Screenshot taken by Selenium after clicking Analyse



Screenshot taken by Selenium after clicking Top words

Bug found by the browser test

evidence/logs/q6-selenium-first-run-bug.log (first run)

```
$ /home/samir/lab-tools/bin/python e2e/test_ui_selenium.py http://flag-on:5000 http://localhost:4444
evidence/screenshots
...
    WebDriverWait(driver, 5).until(result_contains("5 words, 2 unique, 20 characters"))
  File "/home/samir/lab-tools/lib/python3.12/site-packages/selenium/webdriver/support/wait.py", line
138, in until
    raise TimeoutException(message, screen, stacktrace)
selenium.common.exceptions.TimeoutException: Message:
```

The unit tests passed, but the first Selenium run failed: clicking Analyse did nothing. The page script declared `const top`, and `top` is already a read-only global (`window.top`). So the whole script failed with a SyntaxError and no click handlers were attached. I renamed it to `topButton`, and the second run passed all three checks. Only a real browser could catch this bug.

Observations

- The flag toggles behaviour without a rebuild or a code change: **HTTP 404** with the flag off, **HTTP 200** with it on, from the same image.
- One gunicorn worker with 8 threads served about **4,000 req/s** on /healthz and **3,575 req/s** on POST /stats at concurrency 50, with 0 failed requests.

- At concurrency 200, throughput dropped to about 3,000 req/s and p99 rose from 23 ms to 110 ms. The service saturates and requests start to queue, but none fail.
- Grafana's server-side p95 stayed below 0.5 ms during the load, while ApacheBench measured a mean of 12.5 ms per request. The difference is time spent queued in front of the worker, which the in-app histogram cannot see.